

اللغات الصورية ونظرية الآليات

09:00-11:00

18/6/2025

د/ ماجد عسكر

Course code: CS342

3rd Level

Final Exam

Duration: Two hours

This exam for the following program(s) : (Add the program's name(s))

Points: -/50

1

الإسم الرباعي (بالعربي فقط) *

محمد خالد عبدالهادي رماح

2

رقم الجلوس *

2072022149

3

المستوي *

☐ الاول

☐ الثاني

- ☒ الثالث
- ☐ الرابع 2014
- ☐ الرابع 2015
- ☐ الرابع 2016
- ☐ الرابع 2017
- ☐ الرابع 2018
- ☐ الرابع 2019
- ☐ الرابع 2020
- ☐ الرابع 2021

4

البرنامج *

- ☐ عام
- ☐ بايو
- ☐ هندسة
- ☐ ذكاء صناعي
- ☐ الأهلية علوم الحاسب
- ☐ الأهلية نظم المعلومات
- ☒ الأهلية هندسة البرمجيات

5

رقم المعمل *

- ☐ ج*

☐ د۰

☐ ه۰

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☐ اج۱

☒ اد۱

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☐ ا۴

☐ ب۴

☐ ج۴

☐ د۴

☐ ه۴

6

رقم الكمبيوتر *

23

7

الكود * (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

SHVY

☒ Will be reviewed

8

A transition from one state to another may occur without reading a symbol from the input string. This happens in

☒ NFA

☐ DFA

☒ Will be reviewed

9

A finite automaton is deterministic if: from every node there is exactly arc labeled for each character in the alphabet of the language

☐ two

☐ no

☒ one

 Will be reviewed

10

Automata are used to recognize languages, while grammars are used to generate languages.



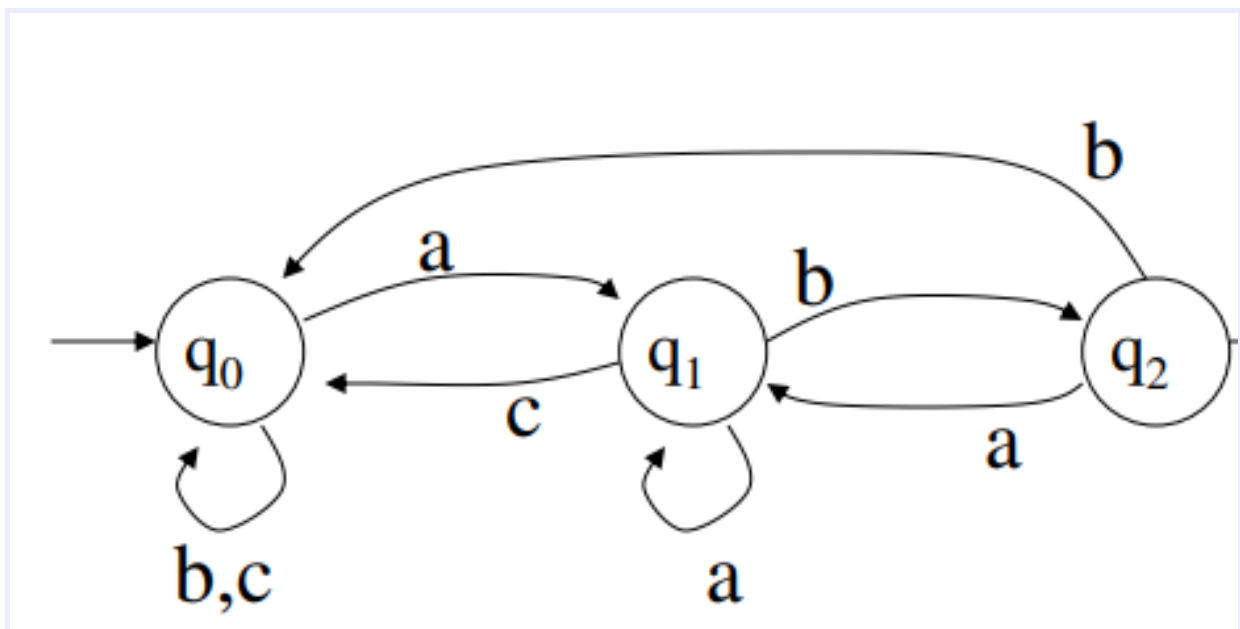
☒ True

☐ False

 Will be reviewed

11

Consider the following DFA, All accepted strings must end in c. 

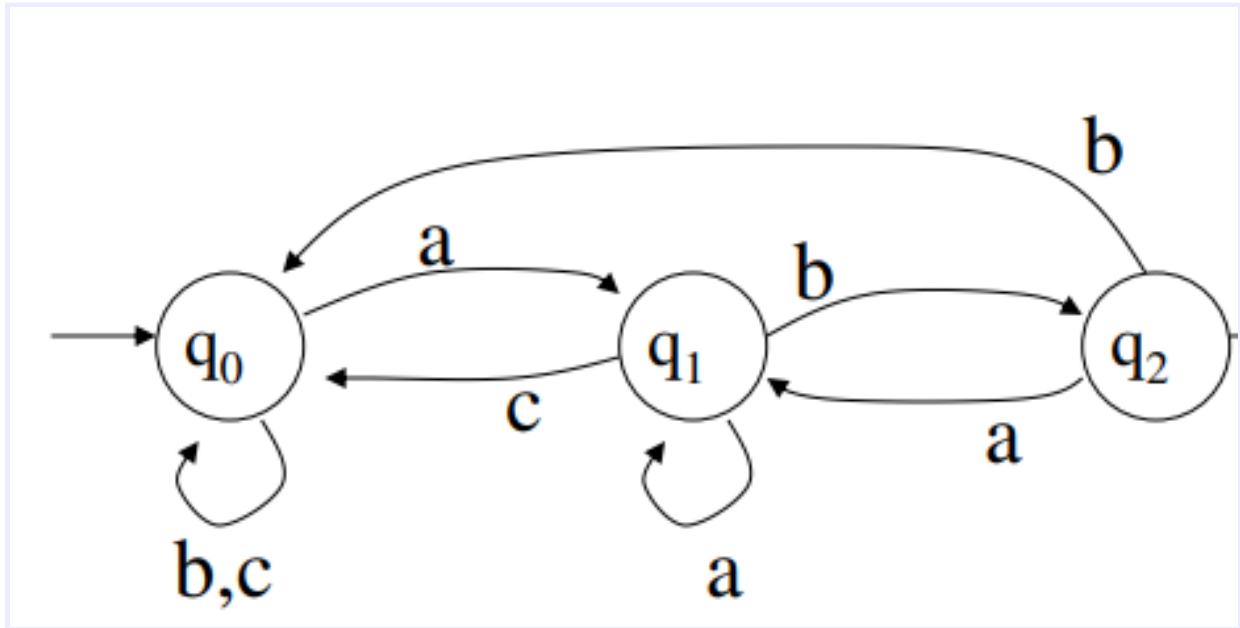


☒ No

☐ Yes

 Will be reviewed

Consider the following DFA, we get to q_2 via ☐



- ☐ (a) path from q_0
- ☐ (b) path from q_0
- ☐ (a) path from q_1
- ☒ (b) path from q_1

☒ Will be reviewed

λ is the shortest string in the language $\{0^i 1^i : i \geq 1\}$

☐

- ☐ True
- ☒ False

☒ Will be reviewed

14

A transition from one state to another may occur without reading a symbol from the input string. This happens in 

☒ NFA

☐ DFA

 **Will be reviewed**

15

In more than one edge with the same label originate from one vertex 

☐ DFA

☒ NFA

 **Will be reviewed**

16

Turing machine has infinite memory, organized as a tape, with a read/write head 

☐ False

☒ True

 **Will be reviewed**

17

Consider the following DFA, Will it accept strings beginning with b or c? 

☒ Yes

☐ No

 Will be reviewed

18

For any finite automata that accepts a given language, where this language has n different strings, the minimum number of states in that FA is



- ☐ greater than or equal to n
- ☐ equal to n
- ☒ less than or equal to n

 Will be reviewed

19

If we have $M = (Q, \Sigma, \partial, q_0, F)$

Σ is



- ☒ finite set of input symbols called alphabet
- ☐ the initial state
- ☐ the transition function
- ☐ a finite, nonempty set of states

 Will be reviewed

20

If we have $M = (Q, \Sigma, \partial, q_0, F)$

∂ is



- ☐ finite set of input symbols called alphabet
- ☐ a set of final or "accepting" states

- ☐ the initial state
- ☒ the transition function

 **Will be reviewed**

21

Will this language automaton accept the string **aba**. $L = \{a^m b^n : m, n \geq 0\}$ 

- ☒ No
- ☐ Yes

 **Will be reviewed**

22

$\delta : Q \times (\Sigma \cup \{\lambda\}) \rightarrow 2^Q$ is a DFA transition function 

- ☐ True
- ☒ False

 **Will be reviewed**

23

Consider the following DFA, Will it accept strings beginning with a? 

- ☒ Yes
- ☐ No

 **Will be reviewed**

24

To show a language is regular, all we have to do is construct a NFA that accepts this language



☐ True

☒ False

Will be reviewed

25

Consider the following DFA, we get to q1 via



☐ an a path from q0 or q1

☐ an b path from q1

☐ an a path from q1

☒ an a path from q0

☐ an b path from q0

☐ an b path from q0 or q1

Will be reviewed

26

The grammar that generates $L = \{a^{n+1} b^{n+1} : n \geq 0\}$ is



☒ False

☐ True

Will be reviewed

27

Let $S = \{1, 2, 3, 4\}$ Then $|S| = ?$ 

☐ 3

☒ 4

☐ 2

☐ 1

 **Will be reviewed**

28

Consider the following DFA, All accepted strings must end in abc. 

☒ Yes

☐ No

 **Will be reviewed**

29

Chomsky Hierarchy is a containment hierarchy of classes of formal grammar 

☐ False

☒ True

 **Will be reviewed**

30

Turing machine has infinite memory, organized as a stack 

☐ True

☒ False

 **Will be reviewed**

31

If we have $M = (Q, \Sigma, \delta, q_0, F)$

Q is 

- ☒ a finite, nonempty set of states
- ☐ the initial state
- ☐ the transition function
- ☐ finite set of input symbols called alphabet

 **Will be reviewed**

32

The next DFA that accepts the formal language $\{a^n b^m\}$, where $n \geq 0$ and $m \geq 1$ 

- ☒ True
- ☐ False

 **Will be reviewed**

33

The computational power of the "automata," is measured in terms of the class of grammar they can recognize 

- ☐ False
- ☒ True

 **Will be reviewed**

34

The extended transition function is represented by: 

- ☐ $\delta : Q \times \Sigma \rightarrow Q$
- ☒ $\delta^* : Q \times \Sigma^* \rightarrow Q$

 **Will be reviewed**

35

Can this DFA be reduced? 

- ☐ False
- ☒ True

 **Will be reviewed**

36

The language L accepted by an NFA can be defined as the language consists of all strings (w) for which there is a walk labeled (w) from the start state to a final state in the transition graph.



- ☒ Right
- ☐ Wrong

 **Will be reviewed**

37

Automata Theory is the study of abstract computing devices, or "machines" 

- ☐ False

☒ True

 **Will be reviewed**

38

Consider the following DFA, All accepted strings must end in bc. 

☒ False

☐ True

 **Will be reviewed**

39

The distinguishable pairs in the next DFA are (q_0, q_4) , (q_1, q_4) , (q_2, q_4) and (q_3, q_4) 

☐ False

☒ True

 **Will be reviewed**

40

Language is a collection of sentences of finite length all constructed from a finite alphabet of symbols 

☐ False

☒ True

 **Will be reviewed**

41

Consider the following DFA, the simplest path to an accepted string is 

☐ bca

☐ acb

☒ abc

 **Will be reviewed**

42

If we have $M = (Q, \Sigma, \delta, q_0, F)$ q_0 is 

☐ a set of final or "accepting" states

☒ the initial state

☐ finite set of input symbols called alphabet

☐ the transition function

 **Will be reviewed**

43

the language generated by this grammar is $L = \{a^n b^{n+1} : n \geq 0\}$



☒ True

☐ False

 **Will be reviewed**

44

Let $S = \{a, b, ab\}$, What is the The number of elements in the powerset of S ? 

☐ 2

☐ 5

☐ 3

☒ 8

 **Will be reviewed**

45

If the NFA has N states, the equivalent DFA can have as many as $2N$ states 

☐ True

☒ False

 **Will be reviewed**

46

consider the list of elements: $A = \{6, 12, 28\}$ then, $A - \{9, 12\} = ?$ 

☐ $\{6, 12, 28\}$

☐ $\{6, 9, 12, 28\}$

☐ $\{12\}$

☒ $\{6, 28\}$

 **Will be reviewed**

47

A is a sequence of transitions from one configuration to another 

☒ computation

☐ grammar

☐ language

 **Will be reviewed**

48

Consider the following DFA, this DFA accept λ ? 

☒ No

☐ Yes

 **Will be reviewed**

49

$\delta : Q \times \Sigma \rightarrow Q$ is a NFA transition function 

☒ False

☐ True

 **Will be reviewed**

50

DFA is useful for proving some theorems. 

☒ Disagree

☐ Agree

 **Will be reviewed**

51

Consider the following DFA, we get to the accepting state from the state 

☐ q1

☒ q2

☐ q3

☐ q0



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